محافظة القاهرة

وحدة تشغيل وإدارة مدارس التكنولوجيا التطبيقية

للتكنولوجيا التطبيقية WE مدرسة الشهيد عمرو مصطفى

Physics

الإسم :

الفصل:

(put (T) for true and (F) for false)

(1) Magnetic flux density measured by weber/m which equivalent to tesla

(2) Magnetic flux (Φ_m) is the total number of magnetic flux lines passing through a surface measure by weber

(3) (θ) in the rule ($\varphi = B\ A\cos(\theta)$) is the angle between the coil plane and the magnetic flux lines.

(4) Magnetic flux density is the total number of magnetic flux lines passing normally through a unit area around the point

(5) **(d) in Ampere's circular law represented any distance between the point of measurement and the wire**

(6) Ampere Right Hand Rule is used to determine the direction of the magnetic field resulting from an electric current in a wire

(7) Permeability measured by (T/m.A)

(8) (Maxwell's Screw Rule) used to determine the direction of the magnetic field resulting from an electric current in a circular coil

(9) Magnetic field due to current passing through a solenoid looks like magnetic field of short bar magnet

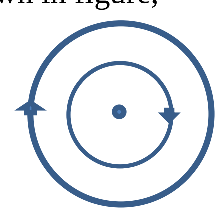
(10) (Soft iron has permeability higher than air) the density of magnetic field produced by a solenoid carrying a current at a point on its interior axis increases when a soft iron bar inserted in it.

(MCQ)

(11) What is the current passes in a wire of length 26.4cm which curved in the form of an arc of a circle of radius 5.6cm to form magnetic flux density of $8.25*10^{-4}$ T at its center.

(a) 0.98 Ampere (b) 98 Ampere (c) 0.89 Ampere (d) 89 Ampere

(12) Two concentric metal rings in one plane, each carries a current of intensity (I) as shown in figure, the direction of the magnetic flux at the common center (m) is:



(a) Rightwards (b) leftwards (c) into the page (d) out of the page

(13) Magnetic lines comes from…………….. looks like long bar magnet

(a) Straight wire (b) Circular coil (c) Solenoid (d) All the previous

(14) We can get the magnitude of magnetic field at the center of circular coil because

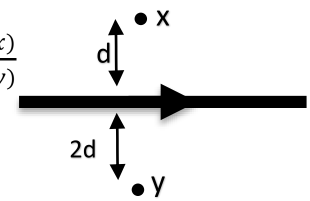
(a) Lines are non-uniform at the center (b) lines are uniform at the center

(c) Lines directed from north to south at the center (d) lines directed from south to north at the center

(15) Rule to determine the direction of the magnetic field resulting from an electric current in a solenoid

(a) Fleming right hand rule (b) Fleming left hand rule (c) polarity (d) Ampere left hand rule

(16) The ratio between the magnetic flux density at point (x) to that at point (y) $\frac{B(x)}{B(y)}$



(a) $\frac{1}{1}$ (b) $\frac{1}{2}$ (c) $\frac{2}{1}$ (d) $\frac{1}{4}$

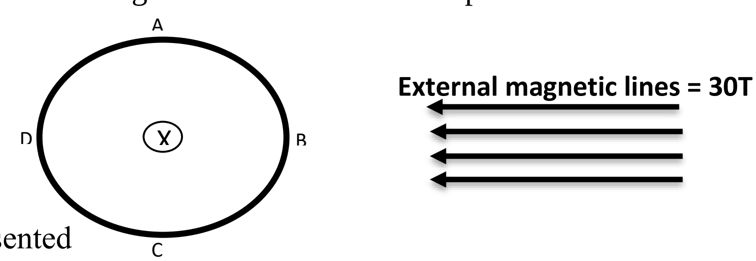
(17) Permeability measured by

(a) T.m/A (b) Ampere (c) wb/m^2 (d) Weber

(18) Magnetic lines can directed from south to north

(a) Outside the magnet (b) inside the magnet (c) near the center (d) Anywhere

(19) - There is a straight wire carrying current to produce magnetic flux density of 30 T on the circumference of the circle ABCD putted in an external magnetic flux of 30T Which point has a resultant magnetic flux of zero T



(a) Point (A) (b) Point (B)

(c) Point (C) (d) Point (D)

(20) (n) in the rule (B = μ n I) for solenoid represented

(a) Number of turns (b) Number of turns per unit length (c) Electric current (d) Permeability

(Write the scientific term)

(21) It is the ability of medium to penetrate the magnetic flux lines ....

(22) it is the total number of magnetic flux lines ....

(23) It is the point at which the total magnetic flux density vanishes ....

(24) It is the total number of magnetic flux lines passing normally through a unit area around the point ....

(25) It is the magnetic flux density when the total number of magnetic flux lines passing normally through a unit area around the point is 1 Wb ....